

Wave Optics – Comprehensive Practice Set

Young's DSE, Fringe Width, Angular Fringe Width, Wavefronts

NEET Pattern – Set 2

Key Formulae:

- Fringe Width: $\beta = \frac{\lambda D}{d}$
 - Angular Fringe Width: $\theta = \frac{\lambda}{d}$ (in radians)
 - n^{th} Bright Fringe: $y_n = \frac{n\lambda D}{d}$
 - n^{th} Dark Fringe: $y_n = \frac{(2n-1)\lambda D}{2d}$
 - Intensity at path difference Δ : $I = 4I_0 \cos^2 \left(\frac{\pi \Delta}{\lambda} \right)$
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Q1. In Young's double-slit experiment, the angular fringe width is 0.05° . If the wavelength of light used is $6000 \text{ \AA}$ and the screen is placed 2 m from the slits, the slit separation is approximately:

- (a) 0.69 mm
- (b) 0.48 mm
- (c) 0.86 mm
- (d) 0.34 mm

Q2. The angular fringe width in a Young's double-slit experiment is 2×10^{-3} radians. When the whole apparatus is immersed in a liquid of refractive index $4/3$, the new angular fringe width becomes:

- (a) $1.5 \times 10^{-3} \text{ rad}$
- (b) $2.67 \times 10^{-3} \text{ rad}$
- (c) $1.33 \times 10^{-3} \text{ rad}$
- (d) $3 \times 10^{-3} \text{ rad}$

Q3. A point source of light produces a wavefront that is:

- (a) Spherical
- (b) Cylindrical
- (c) Plane
- (d) Elliptical

Q4. A linear source of light (like a long straight filament lamp) produces a wavefront that is:

- (a) Cylindrical

- (b) Spherical
- (c) Plane
- (d) Parabolic

Q5. Light coming from a distant star (effectively at infinity) has a wavefront that is:

- (a) Plane
- (b) Spherical
- (c) Cylindrical
- (d) Conical

Q6. In Young's double-slit experiment, the fringe width is β . If the wavelength is increased by 20% and the slit separation is decreased by 25%, the new fringe width becomes:

- (a) 1.6β
- (b) 1.2β
- (c) 0.9β
- (d) 2β

Q7. The ratio of angular fringe width for red light ($\lambda = 750 \text{ nm}$) to that for violet light ($\lambda = 400 \text{ nm}$) in the same YDSE setup is:

- (a) 15 : 8
- (b) 8 : 15
- (c) 3 : 2
- (d) 2 : 1

Q8. In a double-slit experiment, the fringe width is 0.5 mm. The distance between the 3rd bright fringe and the 2nd dark fringe on the same side of the central maximum is:

- (a) 1.75 mm
- (b) 2.0 mm
- (c) 1.5 mm
- (d) 2.5 mm

Q9. Assertion-Reason Type

0.3em]**Assertion (A):** In Young's double-slit experiment, if the slit separation is decreased, the fringe

Q10. Both A and R are true and R is the correct explanation of A

Q11. Both A and R are true but R is NOT the correct explanation of A

Q12. A is true but R is false

Q13. Both A and R are false

Q14. extbfAssertion-Reason Type

0.3em]**Assertion (A):** No interference pattern is detected when two coherent sources are infinitely close.

Q15. Both A and R are true and R is the correct explanation of A

Q16. Both A and R are true but R is NOT the correct explanation of A

Q17. A is true but R is false

Q18. Both A and R are false

Q19. extbfAssertion-Reason Type

0.3em]**Assertion (A):** According to Huygens' principle, no backward wave front is possible. 0.2em]**Reason (R):**

$(1 + \cos \theta)$ where θ is the angle between the ray and the direction of secondary wavelet.

(a) Both A and R are true and R is the correct explanation of A

(b) Both A and R are true but R is NOT the correct explanation of A

(c) A is true but R is false

(d) Both A and R are false

Q20. extbfAssertion-Reason Type

0.3em]**Assertion (A):** When a light wave travels from a rarer to a denser medium, its speed decreases and its wavelength decreases.

Q21. Both A and R are true and R is the correct explanation of A

Q22. Both A and R are true but R is NOT the correct explanation of A

Q23. A is true but R is false

Q24. A is false and R is true

Q25. extbfMatch the Following

0.3em]Match the following :

Column I	Column II
(A) Point source	(P) Plane wavefront
(B) Linear source	(Q) Spherical wavefront
(C) Source at infinity	(R) Cylindrical wavefront
(D) Cylindrical lens with point source	(S) Elliptical wavefront

(a) (A)→(Q), (B)→(R), (C)→(P), (D)→(R)

(b) (A)→(P), (B)→(Q), (C)→(R), (D)→(S)

- (c) (A)→(Q), (B)→(P), (C)→(R), (D)→(S)
 (d) (A)→(R), (B)→(Q), (C)→(P), (D)→(S)

Q27. Match the Following

0.3em] Match the following for a YDSE setup :

Column I	Column II
(A) Fringe width β	(P) $\frac{n\lambda D}{d}$
(B) Angular fringe width	(Q) $\frac{\lambda D}{d}$
(C) n^{th} bright fringe position	(R) $\frac{\lambda}{d}$
(D) n^{th} dark fringe position	(S) $\frac{(2n-1)\lambda D}{2d}$

- (a) (A)→(Q), (B)→(R), (C)→(P), (D)→(S)
 (b) (A)→(P), (B)→(Q), (C)→(R), (D)→(S)
 (c) (A)→(Q), (B)→(R), (C)→(S), (D)→(P)
 (d) (A)→(R), (B)→(Q), (C)→(P), (D)→(S)

Q29. In a Young's double-slit experiment, the intensity at a point where the path difference is $\lambda/3$ is K . If I_0 is the maximum intensity, then K/I_0 equals:

- (a) $3/4$
 (b) $1/2$
 (c) $1/4$
 (d) $1/\sqrt{2}$

Q30. The wavefront is defined as a surface in which:

- (a) All points are in the same phase of vibration
 (b) All points have the same amplitude
 (c) All points have the same intensity
 (d) All points are at the same distance from the source

Answer Key

Q1. (a) 0.69 mm

Q2. (a) 1.5×10^{-3} rad

Q3. (a) Spherical

Q4. (a) Cylindrical

- Q5.** (a) Plane
Q6. (a) 1.6β
Q7. (a) 15 : 8
Q8. (a) 1.75 mm
Q9. (a) Both A and R are true and R is the correct explanation of A
Q10. (a) Both A and R are true and R is the correct explanation of A
Q11. (a) Both A and R are true and R is the correct explanation of A
Q12. (d) A is false and R is true
Q13. (a) $(A) \rightarrow (Q)$, $(B) \rightarrow (R)$, $(C) \rightarrow (P)$, $(D) \rightarrow (R)$
Q14. (a) $(A) \rightarrow (Q)$, $(B) \rightarrow (R)$, $(C) \rightarrow (P)$, $(D) \rightarrow (S)$
Q15. (a) $3/4$
Q16. (a) All points are in the same phase of vibration

All the best for your exam preparation!